

Paper Factory: Evidence-First Orchestration for Agentic Research-Paper Production

Position Paper — System v1.0.0

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<https://github.com/SKZL-AI/paper-factory>

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Abstract

LLM-based agents can now draft scientific manuscripts end to end, but their prose fluency systematically outpaces their evidentiary discipline: claims appear without data, citations resolve to the wrong papers, self-verification closes findings it never checked, and external editing tools mutate text silently. We present *Paper Factory*, a production system built on the opposite premise. A deterministic orchestration core drives a research DAG (P00–P37); LLM harnesses (Claude Code, Kimi, Codex, Pi, OpenCode, LiteLLM-backed workers) are replaceable adapters, never authorities. Every scientific claim must be bound to content-addressed evidence; sixteen global closure invariants (U1–U16) fail closed; an independent verification kernel (VeriHarness/HoH) checks work packages; and the two irreversible steps — final sign-off (P36) and external submission (P37) — remain under human authority by construction. We report evidence from operating the system on a real research paper: 574 automated tests, a 16/16 closure-gate pass, an exact SHA-256 artifact chain from raw data to the arXiv bundle, and 245 writing-assistant suggestions processed capture-only and fully dispositioned. We argue that *evidence-first orchestration*, not better prompting, is the missing layer in agentic science.

1 Introduction: Prose Is Not Evidence

Agentic LLM systems now search literature, run analyses, and write credible manuscripts. What they do not do by default is *prove* anything. Across our own operation of multi-agent research pipelines we repeatedly observed five failure modes:

1. **Unsupported claims.** Final-paper statements with no link to any computed artifact.
2. **Citation misattribution.** References whose DOI resolves — to a *different* paper than the one described.
3. **Vacuous remediation.** Review findings closed by generic post-conditions (“no unsupported claims remain”) that never re-check the concrete finding.

4. **Silent external edits.** Writing assistants (grammar/style tools) rewriting text in ways that shift scientific meaning.
 5. **Premature irreversible release.** Automation drifting toward submission actions that should require a human.
- These are not model-quality problems; they are *systems* problems. A more fluent model produces more fluent unsupported claims. Our position: agentic science needs an orchestration layer in which **evidence is a precondition of prose, verification is independent of generation, and closure is a cryptographic-property-like invariant set — not a summary an agent writes about itself.**

2 Design Principles

Paper Factory implements seven principles:

DP1 — Deterministic core, probabilistic workers.

The pipeline is a fixed DAG (P00–P37) executed by a deterministic core with persistent, append-only state. LLM harnesses are interchangeable workers behind a provider router; the backend model identity is recorded separately from the harness name.

DP2 — Evidence before prose.

Artifacts carry authority tiers T0–T4 (raw data and registries down to drafts and chats). No empirical claim enters the paper without evidence linkage.

DP3 — Claim–evidence binding.

Claims form a graph bound to concrete artifacts — metrics, tables, analyses — each content-addressed by SHA-256. Numbers in prose are generated from metrics via macros, never hand-typed.

DP4 — Independent verification.

Work packages are verified by a separate kernel (VeriHarness/HoH: plan → develop → independent QA → receipts → accept/reject) integrated through an adapter. The generator never grades itself.

DP5 — Fail-closed gates.

Sixteen global closure invariants (U1–U16) audit claim support, citation identity, number-to-metric binding, remediation integrity, and external-edit reconciliation. Exit code 0 means CLOSED and nothing else; insufficient evi-

Table 1: Operational evidence at the v1.0.0 freeze.

Measure	Result
Automated tests	574 passed, 2 skips
Pipeline nodes P21–P36	all PASS
Closure invariants U1–U16	16/16 PASS
Node summary	33 / 4 / 0
Citations verified	21/21
Writing-assistant suggestions (unique)	245
applied under semantic guards	75
rejected with evidence	124
not applicable	46
undispositioned	0
Artifact hash chain end-to-end	MATCH
P37 external submission	NOT RUN

dence yields DEGRADED/FAILED, never a fabricated pass.

DP6 — Human authority at irreversible steps.

P36 is an explicit, receipt-bound human sign-off. P37 (external submission) is never executed automatically.

DP7 — Append-only receipts.

Every transition emits receipts; corrections amend history instead of rewriting it. Reviews are adversarial by design (dual independent reviewers, instructed to refute).

3 Architecture

Figure 1 shows the layering. The core (this system) is a Python CLI — `paper-factory complete` runs from a plain shell with no interactive assistant involved. State lives in SQLite plus immutable JSON/JSONL/YAML artifacts; every node of the DAG reads and writes content-addressed evidence.

Three boundaries are load-bearing. (i) *Adapter boundary*: harnesses are workers; swapping Claude Code for Kimi changes labor, not truth. (ii) *Verification boundary*: VeriHarness/HoH decides accept/reject on receipts; Paper Factory cannot override a rejection. (iii) *Authority boundary*: a provenance firewall decides which process may write protected paths (e.g. final prose); unknown backends are denied by default, and external edits (e.g. a writing assistant operating on a DOCX) are reconciled against the frozen canonical manuscript via claim, number, citation, and math diffs before they can influence the release.

4 Evidence from Operation

Paper Factory v1.0.0 was frozen only after a real end-to-end pilot: a mass-invariance research paper taken from draft-assisted intake through analysis, citation verification, external writing-assistant review (Microsoft Word + Paperpal, driven ownership-scoped and capture-only), adversarial dual review, and arXiv-bundle packaging. Table 1 summarizes.

Two observations matter more than the pass counts. First, hardening *increased* honest failures during development:

making remediation finding-specific (killing vacuous resolution) and binding numbers to named metrics surfaced real defects that summary-level checks had hidden. A system that never fails is not strict; it is unaudited. Second, all 124 rejected writing-assistant suggestions were rejected *with evidence*, and none of the 245 unique suggestions touched the manuscript silently — external-edit reconciliation (U16) is a gate, not a diff nobody reads.

4.1 The Closure Invariants

For concreteness, the sixteen invariants every run must satisfy verbatim:

- U1. Every primary claim has evidence.
- U2. Every scientific number matches generated source.
- U3. Figures/tables match their manifests.
- U4. Citations resolve and support the contextual claim.
- U5. No unresolved CRITICAL/MAJOR review finding.
- U6. Reviewed manuscript = release manuscript.
- U7. Code/data commits match manifests.
- U8. No secrets or private transcripts enter the release.
- U9. Writing-assistant state is PASS or explicit HUMAN_REQUIRED.
- U10. The clean bundle independently rebuilds.
- U11. Final-prose files have allowed origin receipts.
- U12. Forbidden model families did not modify protected prose.
- U13. Backend identity is known for final writer invocations.
- U14. Marking status is reported honestly.
- U15. Reviewer replacement prose is not silently copied.
- U16. External writing-assistant edits underwent semantic reconciliation.

None of these is a prompt. Each is an executable check over artifacts, receipts, and hashes, evaluated by the core at closure time — an agent cannot argue its way past them.

5 Limitations and Open Problems

We are deliberately not claiming generality. (i) Evidence comes from one real pilot family plus synthetic suites; broader intake modes (data-only, code-only) need more pilots. (ii) Semantic-diff artifacts are partially assertion-level markers, not deep proofs. (iii) External-tool automation (e.g. a writing assistant’s web pane) is inherently UI-fragile; we mitigate with capture-only semantics and ownership-scoped process control, but drift remains an operational risk. (iv) The verification kernel and runtime have their own maturity limits; Paper Factory records these honestly as DEGRADED rather than hiding them. (v) Exit-code discipline and closure invariants resist, but cannot eliminate, adversarial or confused *human* overrides; durable human decisions are therefore append-only and collision-hardened, yet still human.

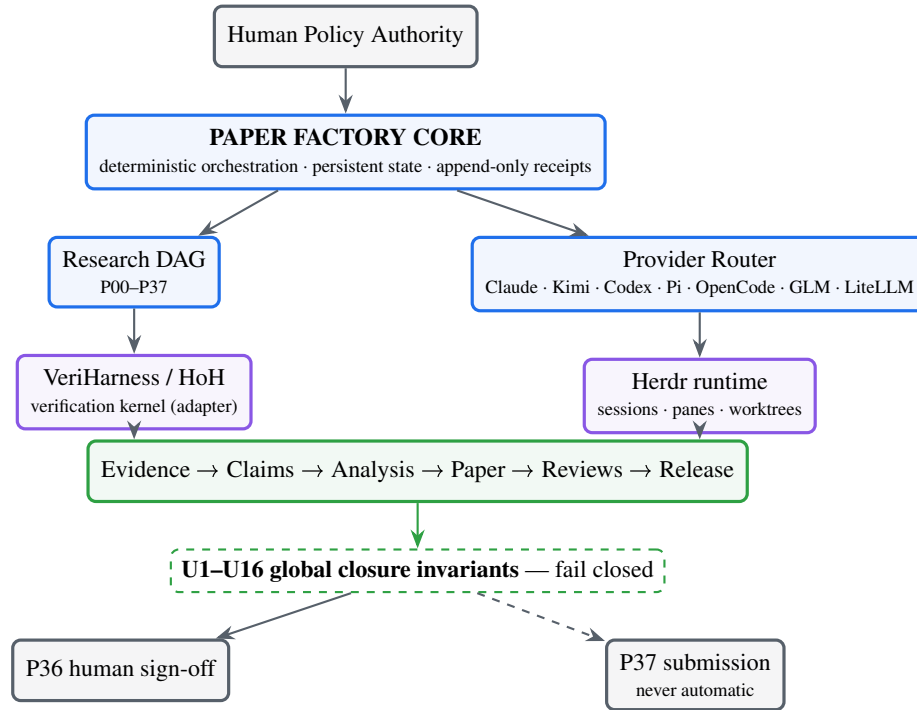


Figure 1: Paper Factory architecture. The deterministic core owns *what* must happen; the verification kernel owns *whether* it happened; the runtime owns sessions and isolation; the human owns irreversible steps.

6 Related Work

The system builds on established infrastructure rather than replacing it: large language models as workers [1]; community scholarly-metadata registries for citation identity (Crossref [2], OpenAlex [3]); and W3C PROV as the conceptual basis for artifact provenance [4]. VeriHarness/HoH contributes the harness-of-harnesses verification model with receipts and pre-registered gates [5]. Relative to agentic writing frameworks, Paper Factory’s distinction is negative as much as positive: it is designed around everything an agent must *not* be trusted to do — grade itself, close findings vacuously, edit prose silently, or submit irreversibly.

7 Conclusion and Call to Action

Agentic science will be believed exactly to the extent that its artifacts are auditable. We offer Paper Factory as one concrete, operating answer: deterministic orchestration, evidence-bound claims, independent verification, fail-closed closure invariants, and humans at the irreversible steps. The system, its test suite, its freeze evidence, and its honest list of limitations are public. We invite the community to attack the invariants — that is what they are for.

References

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